# RealQM for $H_2$ and He

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**Abstract**

RealQM replaces the many-electron wavefunction by $N$ one-electron charge densities on non-overlapping domains, interacting through Coulomb's law with each pair counted once. At one electron it is the minimal model that Coulomb's law admits—a kinetic penalty quadratic in $\nabla\Psi$ against the kernel attraction, nothing else. Its *form* is forced, but the coefficient of the kinetic term is a scale the model cannot derive, and hydrogen's observed spectrum is what fixes it. Hydrogen therefore *calibrates*—the energy scale of the whole construction, and separately the one constant the discrete scheme needs—and cannot test anything.

$H_2$ is that same model extended to two electrons with *no further calibration*, in the two forms the interface condition admits: imposed strongly on a sharp partition, or weakly on a diffuse one. Both are predictions in the strict sense—nothing has been fitted at the two-electron level—and both agree fairly with observation, each on one observable: the strong form places the bond within 3.7%, the weak form the binding energy within 7.7%, while each is substantially worse on the other, and the two straddle the observed binding energy from either side. Helium is reported for the strong form only: the weak form returns a state that fails the sphericity test, for an implementation reason we identify, and we decline to quote an energy from it. The model acquires content at two electrons, where the densities must be partitioned and must meet, and we examine the two smallest such systems: $H_2$, whose interface lies between the kernels, and He, whose interface passes through one. Two defensible interface treatments—strict homogeneous Neumann on a sharp partition, and amplitude continuity enforced weakly on a diffuse one—*bracket* the observed energies of both systems from opposite sides. We argue that this interval, and not either endpoint, is what the construction predicts, and that the width of the interval is the honest statement of its accuracy at two electrons.

## 1 The model

The unknown is a *single* real function $\Psi \in H^1(\mathbb{R}^3)$ together with a partition of space into $N$ disjoint domains $\Omega_1, \ldots, \Omega_N$ filling $\mathbb{R}^3$. Writing $\psi_i = \Psi|_{\Omega_i}$, so that

$$\Psi \;=\; \sum_i \psi_i, \qquad \int_{\Omega_i} \Psi^2 = 1 \;\text{ for each } i, \tag{1}$$

the energy is

$$E[\Psi, \{\Omega_i\}] \;=\; \tfrac{1}{2}\int |\nabla\Psi|^2 \;-\; \int K\,\Psi^2 \;+\; \sum_{i<j} \iint \frac{\psi_i^2(\mathbf{x})\,\psi_j^2(\mathbf{y})}{|\mathbf{x}-\mathbf{y}|} \;+\; \sum_{a<b} \frac{Z_a Z_b}{R_{ab}}, \tag{2}$$

with $K = \sum_a Z_a/|\mathbf{x}-\mathbf{R}_a|$, and the problem is to *minimise $E$ over both $\Psi$ and the partition.* The domains being disjoint, $\Psi^2 = \sum_i \psi_i^2$ with no cross terms, so the third sum is a genuine pair interaction counted once, and no electron interacts with itself.

Three features of this statement are worth drawing out, because the rest of the paper turns on them.

**Continuity is not imposed; it is the function space.** $\Psi$ is required to lie in $H^1$, so it is continuous across every interface by assumption, and no matching condition is added by hand. What *is* free is where the interfaces are.

**The interface condition is a consequence, not an input.** Varying $\Psi$ at a *fixed* partition leaves a boundary term $\oint \psi_i \, \partial\psi_i/\partial n$ on each $\partial\Omega_i$, so the natural condition is $\partial\Psi/\partial n = 0$ from each side—nothing is imposed. Solved that way the two one-sided values at an interface generically disagree, and the interface is then placed where they agree, which is what varying the *partition* supplies. Three conditions, one extra unknown; the count closes. The two implementations in §3 are two discretisations of this one variational problem and differ only in how they represent $\Psi$ near the interfaces—not in the problem being solved.

**There is one function, not $N$ of them.** The decomposition (1) is bookkeeping: $\psi_i$ is a restriction of $\Psi$, not an independent orbital, and the $\Omega_i$ are the only thing distinguishing one electron from another. The dimension of the unknown is three regardless of $N$, which is what makes the model computable and therefore testable at all.

**Two mechanisms, doing two different jobs.** It is worth separating them, because they are easily run together. The Coulomb term does *not* keep two electrons apart: for smooth densities $\iint \rho(\mathbf{x})\rho(\mathbf{y})/|\mathbf{x}-\mathbf{y}|$ is finite even when the two coincide, so it permits coincidence at a finite price and cannot forbid it. What forbids it is the *partition*, which is a constraint and not a penalty: the domains are disjoint by construction, so overlap is unavailable rather than expensive. What the Coulomb term does is resist *compression*—as two domains are pushed together $V_{ee}$ rises, and that sets where the interface sits and how density distributes inside each domain.

The distinction matters for what follows. Between *equivalent* domains across a mirror plane the partition contributes nothing beyond what Coulomb already supplies: the sharp and continuous constructions of §3 agree there to $4 \times 10^{-8}$. So in exactly that configuration the construction's whole resistance to compression is the classical Coulomb repulsion of two charge clouds, and the over-binding reported in §5 is the measurement that this alone is not enough. $H_2$ at its midplane and He at its split plane are both that configuration, which is why they are the systems to examine.

**What is new, and what is missing.** Relative to one electron the construction adds exactly two things: the densities do not overlap, and they must meet somewhere. It also *subtracts* one: it fixes what resists their approach, and the answer is set out below—the partition makes overlap impossible while the Coulomb term alone resists compression. That is a property of the model, stated in its own terms and checkable within it, and it is why no two-electron system should be expected to come out exactly.

## 2 One calibration, on hydrogen

At one electron (2) is $\frac{\kappa}{2}\int |\nabla\Psi|^2 - \int (Z/r)\Psi^2$ minimised subject to $\int \Psi^2 = 1$. A kinetic penalty quadratic in $\nabla\Psi$ is the lowest-order rotation-invariant term that prevents collapse and no simpler functional binds at all, so the *form* is forced; but the coefficient $\kappa$ is not. It is a scale, and the model contains nothing from which to derive it.

**The coefficient of the kinetic term is the model's one physical calibration, and hydrogen sets it.** Requiring the minimiser to reproduce hydrogen's observed spectrum fixes $\kappa = 1$ in atomic units, i.e. the familiar $\frac{1}{2}\int |\nabla\Psi|^2$, whereupon the minimiser is $\Psi = \sqrt{Z^3/\pi}\, e^{-Zr}$

with energy $-Z^2/2$. We are explicit that this is a calibration and not a derivation. The construction contains *one* kinetic coefficient and nothing from which to compute it: no mass appears anywhere in (2), and no constant of action, so there is no product of such quantities to be supplied from elsewhere and no separation of $\kappa$ into factors that the model could make. Hydrogen's spectrum fixes the single number, and in doing so sets the energy scale of everything that follows. That is the sense in which the one-electron model is *calibrated* rather than predictive, and it is why no agreement at one electron can be evidence for the construction.

With $\kappa$ fixed, the continuum model is complete and contains no further adjustable quantity. Its discrete realisation needs one more constant, of a different kind: a grid cannot represent $1/r$, so the kernel carries a softening, $K = Z/\sqrt{r^2+\varepsilon^2}$ with $\varepsilon = \sigma h$. We fix $\sigma$ once, by requiring the one-electron answer to be exact. In the axisymmetric solver used below this gives

$$\sigma = 0.2846 \;: \qquad E(\mathrm{H}) = -0.500227 \quad (0.045\%), \tag{3}$$

and in the three-dimensional solver $\sigma = 0.627$, read off a four-point ladder ($-0.508456$, $-0.500962$, $-0.493833$, $-0.486591$ at $\sigma = 0.4, 0.6, 0.8, 1.0$, linear in $\sigma$ to better than a percent) and then verified at the production grid, where it returns $E(\mathrm{H}) = -0.499901$, an error of 0.02%. The two values differ because the two codes discretise differently; each is fixed by its own one-electron exactness, which is the point.

That same one-electron row carries a null test worth stating, because it costs nothing. A single domain has no interface, so any interface measure evaluated on it must vanish; the quantity $\int u^2 |\nabla w|\, dV$, which is the regularised form of $\oint_\Gamma u^2 dS$ used in §5, returns $8 \times 10^{-4}$ here against 0.417 for $H_2$. A measure that did not vanish on one electron would be measuring something other than an interface.

**The mesh is declared, and $\sigma$ is a function of it.** Two things must be separated here. The spacing $h = 0.1\, a_0$ is *declared*: it is fitted to nothing, chosen once, and used for every system in this paper. The softening is then *determined* rather than chosen—at each $h$, the condition $E(\mathrm{H}) = -1/2$ fixes it, and at $h = 0.1$ that gives $\sigma = 0.627$. So the discrete scheme carries one declared number and one determined one, and strictly the determined one is $\sigma(h)$: a different mesh requires re-solving the same condition, not reusing 0.627. As $h$ decreases the grid resolves the kernel better and less softening is needed, so $\varepsilon = \sigma(h)\, h \to 0$—the scheme does not freeze a finite softening into the continuum limit.

What this does *not* claim is that $h = 0.1$ is converged. It is not, and the paper says so where it matters: the weak construction's bond error falls $34.9 \to 28.3 \to 21.4\%$ at $h = 0.1395, 0.10, 0.08$ and is still falling. Every number below is therefore reported at a declared mesh and not at a limit, which is the honest status of a model whose results depend on $h$ and whose $h$-dependence is measured rather than assumed away.

This is the second constant, and the last. Three properties of it matter. It is performed on a *one-electron* system, where the model has no content of its own and the reference is exact, so no two-electron observable is consumed. It is a single number, not a function of anything. And it is fixed before any two-electron system is computed. Everything below follows with no further calibration.

**What this is not.** It is not an optimisation of $\varepsilon$ against a residual. That operation is available and must be refused: the total force residual is monotone in $\varepsilon$ and minimised by unphysical settings—hydrogen ceases to bind at $\varepsilon = 0.75h$—so a scheme that tunes $\varepsilon$ to improve an error measure drifts away from the physics it is meant to describe. Fixing it on a known energy is a different operation and has a unique answer.

## 3 The interface, and the two constructions

For a fixed partition, free variation of each density in (2) leaves a boundary term $\oint u_m \, \partial u_m/\partial n$, so the natural condition is homogeneous Neumann, $\partial u/\partial n = 0$, on each domain's own boundary—nothing is imposed. Solved that way the amplitudes at the interface generically *disagree*, and the interface is then placed where they agree. Continuity is therefore not a second assumption about $\Psi$; it is the equation that determines $\Gamma$. Three conditions, one extra unknown.

Two implementations each supply one half of that scheme and take the other half from elsewhere.

**(A) Sharp partition, strict Neumann.** Labelled cells, with the kinetic flux through any face joining cells of different ownership *omitted at assembly*. Omitting a face *is* the homogeneous-Neumann condition—zero flux implemented by not adding a flux, with no extra equation. This imposes the condition exactly but requires $\Gamma$ to be known, which for $H_2$ and He is supplied by symmetry. One trap is worth recording: taking a *submatrix* of the full stiffness instead retains the diagonal contributions of faces leading into the excluded region and so imposes Dirichlet, pushing density off the boundary and under-binding. Measured on a one-electron problem outside a core, 3.977 against 6.177 eV; the latter matches a one-dimensional radial solve to 2%.

**(B) Diffuse partition, continuity enforced weakly.** An occupancy $w_m$ evolves on the local amplitude mismatch $c = u_m - \max_{n \neq m} u_n$ and converges where that mismatch vanishes, so this construction genuinely *finds* $\Gamma$—the only one of the two that does. Its Neumann condition is only natural, not imposed: varying $\frac{1}{2}\int w|\nabla u|^2$ makes $w\,\partial u/\partial n = 0$ stationary, but on a finite grid a face straddling the interface carries half flux rather than none. That half-flux is an extra kinetic cost sitting exactly at the interface, and it is a first-order discretisation error, not a structural one: $H_2$'s bond error falls $34.9 \to 28.3 \to 21.4\%$ at $h = 0.1395, 0.10, 0.08$ and is still falling.

**The two are not a choice between theories.** They are two incomplete implementations of one: (A) imposes the condition and is told where $\Gamma$ is; (B) finds $\Gamma$ and imposes the condition weakly. Combining them—binary faces on a *diffuse* occupancy—does not work and should not be attempted: it gives $H_2$ a binding energy of +110% and a dipole of 0.108–0.187 D where reflection symmetry requires identically zero, because closed faces make an amplitude jump at the interface free. The diagnostic is direct: the kinetic energy of the assembled field $\Psi = \sum_m w_m u_m$, which charges that jump, exceeds the per-domain sum by a factor of 8.07, so the discrete solution is not in $H^1$.

## 4 $H_2$: two kernels, interface between them

Hydrogen fixed one constant and settled nothing else. $H_2$ is the same model carried to two electrons *without any further calibration*: the same $\sigma$, the same functional, the same $i \neq j$ Coulomb rule, and the measured separation as the only other input. Every number in this section is therefore a prediction in the strict sense, and that is the only reason the agreement below is worth reporting—a two-electron constant fitted here would have made it worthless.

It is also the mildest two-electron system available. Each kernel is $Z = 1$, so the softening calibrated on hydrogen applies unchanged to both and nothing is transferred from a different kernel charge; and the interface sits at the midplane by reflection symmetry, in the *tail* of both densities rather than near either nucleus. The extension admits exactly two forms, differing only in how the interface condition is imposed—strongly, on a sharp partition, or weakly, on a diffuse one—and we report both.

With $\sigma$ fixed by (3) and the atomic reference measured at the *same* $\sigma$—a requirement, not a nicety, see below—the binding energy comes out

$$E_{\mathrm{bind}} = 2E(\mathrm{H}) - E(\mathrm{H_2}) = 2(-0.499901) - (-1.160855) = 0.161053,$$

against $D_e = 0.174470$: **under-binding by** 7.7%, at $R = 1.5\,a_0$ on the commensurate grid nearest the observed 1.40028. The sharp construction at its own calibration gives +59.6% at its own minimum while placing the bond within −3.7%. The two therefore fall on *opposite sides* of the observed binding energy.

**The bond, which is the better-behaved observable.** The separation at which each construction's energy is lowest is a prediction carrying no reference at all beyond the measured $R_e = 1.40028$. The sharp construction puts it at $1.349\,a_0$, −3.7%, from a parabola through three commensurate points about the minimum. The weak construction at $\sigma = 1$ gives $1.8 \pm 0.1\,a_0$ (+28%), the uncertainty being the width of the band within which the energy differences fall below the $\pm 0.0018\,E_h$ initialisation scatter. At the calibrated $\sigma = 0.627$ its minimum has not yet been located: a sharper kernel deepens the well and should shorten the bond, so we expect it between those two figures, but we have one point and not a scan, and report the gap rather than interpolate it.

The ordering is worth noting because it is the opposite of the energy's. On the bond the sharp construction is the accurate one, within four percent; on the binding energy it is the inaccurate one, at +59.6% against the weak split's −7.7%. No single construction here is better on both.

**Two implementations, complementary errors, and what that does and does not license.** The two differ in *exactly one* respect—how the interface condition is imposed—and everything else about them is the same model, the same calibration and the same geometry. So the complementarity is not a coincidence of two unrelated methods: the weak construction's error is a half-flux localised *at the interface*, which for a diatomic is precisely where the bond length is determined, and it is therefore large in the bond and much smaller in the binding energy. That the pair gets one observable right each is a statement about where the deficiency lies, and it localises it to the interface treatment alone.

We are careful about what this licenses. Quoting the sharp bond together with the weak binding as "the model's answer" would require a rule, settled in advance, for which construction to trust for which observable. For the bond there is such a rule: $\mathrm{H_2}$'s interface position is fixed by reflection symmetry, so the sharp construction's *only* handicap—that $\Gamma$ must be supplied to it from outside—is inactive here, and its geometry is accordingly the one to quote. For the binding energy no corresponding argument is available: the weak construction's handicap is never inactive, so its −7.7% is not privileged over the sharp +59.6% by anything beyond being nearer the measurement, and selecting it on that ground would be fitting. The pair therefore *diagnoses*; it does not predict both observables at once, and we do not present it as doing so.

**A warning earned the hard way.** The reference must be computed at the same $\sigma$ as the molecule. Paired instead with an atom measured at $\sigma = 1$, the same $\mathrm{H_2}$ run reports $E_{\mathrm{bind}} = 0.188011$, i.e. +7.8% and apparently *over*-binding—the sign inverted by a mismatched reference. Since $\sigma$ sets the cusp resolution and hence every energy, a reference at a different $\sigma$ is not a reference at all.

## 5 He: hard split and weak split

Helium is the same two-electron question asked under far worse conditions. The kernel is $Z = 2$, so the softening calibrated on hydrogen is being extrapolated—though not unreasonably: the

one-electron ion $He^+$ at the same $\sigma$ gives $-2.053249$ against the exact $-2.0$, only 2.66% deep, so the cusp itself is well handled. And the interface passes *through* the kernel rather than lying between two of them, so it cuts the density maximum: $\int_\Gamma u^2 dS = 1.681$ against $H_2$'s 0.496, a factor of 3.4.

**The hard split.** With $\Gamma$ the plane through the kernel and the condition imposed exactly, $E(\mathrm{He}) = -3.026749$ against the observed $-2.903724$: **over-binding by** 4.2%. That is the one helium figure this paper reports.

**The weak split is not yet measurable, and we report that rather than a number.** Every attempt returns a state that fails the sphericity test: a computed dipole of 0.514 D where symmetry requires identically zero. By the standard set out in §7—a symmetric system's dipole is a pure error measurement needing no reference, and is the one check that cannot be argued with—such a state is a failed computation and not a result, so its energy is not quoted here.

The cause has been identified and it is an implementation defect rather than a property of the construction. At the kernel's own cell the relative coordinate vanishes, so the test that assigns cells to hemispheres ($\mathbf{r} \cdot \hat{\mathbf{a}} < 0$ against $\geq 0$) excludes one sibling from that cell and grants it to the other. The softened attraction there is $Z/\varepsilon \approx 32\, E_h$ per unit density, so the two hemispheres are inequivalent *by construction*. Two controls support this and no other hypothesis: releasing the per-step enforcement reproduces the state *bit-identically*, so the asymmetry is fixed at initialisation; and rotating the dividing plane to a diagonal moves the dipole by only 4.7%, so it is not an alignment of the plane with the grid. Neither control changes the kernel cell, which is why neither helps.

**What helium does establish, independently of that defect.** The one-electron ion on the same kernel at the same $\sigma$ gives $-2.053249$ against the exact $-2.0$, 2.66% deep, with $T = 1.8855$ against 2.0. So the $Z = 2$ cusp and the transferred softening are *both* sound, and whatever limits helium is a two-electron matter. That is worth having: it rules out the calibration and the kernel as explanations before any two-electron number is in hand.

And the geometry is the reason to expect helium to be the harder case either way. Its interface passes through the kernel rather than between two of them, so it cuts the density maximum: $\int_\Gamma u^2 dS = 1.681$ against $H_2$'s 0.496. Whatever error an interface treatment carries is weighted there by the largest density in the system.

## 6 $HeH^+$: the asymmetric case

Both systems above have interfaces fixed by symmetry, where the two constructions provably coincide, so neither can settle which formulation is right. The smallest two-electron system that can is $HeH^+$: two electrons on kernels of charge 2 and 1, net charge +1, equilibrium separation $R_e = 1.4632\, a_0$. Its interface lies at no symmetry plane, so amplitude matching and zero flux are genuinely different conditions there and the $\sim$5-fold discrepancy they show at a displaced interface should appear in an observable.

The construction is the natural one and needs nothing new: two domains, one electron each, one carrying a $Z = 2$ kernel and the other $Z = 1$. There is no pseudopotential, no core, no split ambiguity, and $r_c = 0$ throughout.

**One confound must be declared, and it is not small.** The two kernels have *different* charges, so a single softening $\sigma$ cannot be calibrated for both: the value fixed on hydrogen is not the value a $Z = 2$ cusp requires, and the sharper cusp is the less well resolved. An error in $HeH^+$ is therefore attributable either to the interface asymmetry—the thing we want

to measure—or to the $Z$ mismatch, and the two are not separated by this system alone. The honest reading is that $HeH^+$ tests the formulation only to the extent that the $\sigma$ transfer error is smaller than the effect sought, which has not been established.

**The clean alternative, with equal kernels.** The confound disappears if the asymmetry is put in the interface rather than in the nuclei: $H_2$ at fixed $D$ with the dividing plane *displaced* off-centre, each domain still carrying unit charge. Both kernels are then $Z = 1$, one $\sigma$ serves both, and the only asymmetry is the one under test. That measurement exists. Writing $\delta E = \frac{1}{2}kz_i^2$ for a displacement $z_i$,

| $h$ | $k$, continuity enforced | $k$, strict per-domain | ratio |
|---|---|---|---|
| 0.250 | 0.7359 | 0.1296 | 5.68 |
| 0.125 | 0.7394 | 0.1472 | 5.02 |

The continuity stiffness is converged—0.5% across a halving—while the strict one is not, climbing 13.6% and still moving. That asymmetry is the sharper statement: the strict construction charges for a *discontinuity*, a mesh-scale object, so it has no well-defined interface stiffness in the continuum limit, whereas the continuity construction does. For an interface that must *move*—which is every molecule without a symmetry plane—that is a difference in kind rather than in accuracy.

**So the programme is: $HeH^+$ for the observable, displaced $H_2$ for the mechanism.** $HeH^+$ gives a real system with a real reference and a genuinely asymmetric interface, at the cost of the $\sigma$ confound; displaced $H_2$ isolates the mechanism exactly but is not an observable configuration. Neither alone decides it; together they do, and the prediction to test is that the construction which converges in interface stiffness is the one whose geometry converges in $HeH^+$.

## 7 What this establishes, and what it does not

**Not that the model is accurate at two electrons.** It is not. On $H_2$ the two forms disagree by sixty-odd points of $D_e$, and on helium only one of them yields a state that passes the sphericity test at all. That width is the expected consequence of a declared property of the construction—the partition makes overlap impossible while the Coulomb term alone resists compression, and between equivalent domains across a mirror plane the partition contributes nothing—rather than a numerical failing to be driven down. The testable claim is that the error is *systematic*: one sign, one scale, and cancelling in differences taken at fixed composition.

**And not that either observable has been predicted twice.** One form places $H_2$'s bond within 3.7% and the other its binding energy within 7.7%, from a single calibration performed on one electron. That is the paper's positive result, and its limit is stated in §4: the pair localises the deficiency to the interface treatment, and only the bond comes with a reason—$H_2$'s $\Gamma$ being fixed by symmetry—for preferring the form that gets it right.

**A caveat on the references.** The figure $-1.174470$ for $H_2$ is itself a *computed* quantity rather than a measured one, and the total electronic energy is in any case not an observable. The observables here are the dissociation energy, the bond length, and a dipole that must vanish by symmetry. We report total energies because they are what the two constructions differ in, not because they are measurable.

**What the dipole buys.** For a symmetric system the computed dipole must be identically zero, so any nonzero value is a pure error measurement needing no reference of any kind. It is the one check in this paper that cannot be argued with, and it is what disqualified the combined construction of §3 and would disqualify any future variant that improved the energy while losing the symmetry.